

Paper GR-NewtonRecovery

$$\ell_{\text{ED}}^2 = \ell_P^2 \text{ (Planck Identification)}$$

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of the Planck-length identification $\ell_{\text{ED}}^2 = \ell_P^2$ from nothing. Result is conditional on the 13 ED primitives (Paper_087) + declared paper-specific postulates.
2. It does **not** claim closed-proof reconstruction in the Hardy 2001 / CDP 2011 / Coecke-Kissinger operational-reconstruction sense. ED sits in the substrate-ontology lineage ('t Hooft cellular-automaton, Wolfram Ruliad, causal-set program), not the operational-reconstruction lineage.
3. It does **not** override Paper_095's three-tier verdict grammar. Verdict for this paper is stated in §1.
4. It does **not** claim independent derivation of the Planck length from substrate parameters — the identification $\ell_{\text{ED}} = \ell_P$ is dimensional rearrangement of $G, \hbar, c$ via Paper_027, not a first-principles substrate computation.
5. **INHERITED vs FORCED breakdown:** Newton's $1/r^2$ form is FORM-FORCED at substrate level (Paper_027); the numerical value of G (and hence of ℓ_P) is VALUE-INHERITED via empirical matching.

Abstract

This paper supplies the substrate-level identification $\ell_{\text{ED}}^2 = \ell_P^2 = \hbar G / c^3$, fixing the P08 substrate scale to the Planck length. The result is forced by joint requirement of (i) Newton-recovery at substrate level (Paper_027 supplying $G = c^3 \ell_{\text{ED}}^2 / \hbar$), (ii) V1 long-range kernel structure (Paper_089), and (iii) empirical matching to the measured value of G . The structural identification is FORM-FORCED; the numerical value $\ell_P \approx 1.6 \times 10^{-35}$ m is VALUE-INHERITED via empirical $G, \hbar, c$. Per Paper_095 form-FORCED / value-INHERITED methodology, verdict is **M3**. Key falsifier **F1**: any substrate-level derivation yielding $\ell_{\text{ED}} \neq \ell_P$ would refute the Newton-recovery identification.

1 Statement of Result

Claim. The substrate scale ℓ_{ED} (P08) is identified with the **Planck length** ℓ_P in the substrate-gravity arc:

$$\ell_{\text{ED}}^2 = \ell_P^2 = \frac{\hbar G}{c^3}.$$

This identification is **forced** by the requirement that the substrate-gravity arc recover Newton's law $a = GM/r^2$ in the appropriate long-range / weak-field limit, combined with the substrate-derived expression $G = c^3 \ell_{\text{ED}}^2 / \hbar$ (Paper_027) and the V1 kernel's long-range structure (Paper_089).

The structural identification $\ell_{\text{ED}} = \ell_P$ is FORM-FORCED. The numerical value of ℓ_P (and equivalently the value of G) is INHERITED via empirical matching.

Verdict: **M3** — Newton-recovery is form-FORCED at substrate level (Paper_027); the Planck-scale identification is value-INHERITED via dimensional rearrangement of $G, \hbar, c$.

2 Primitive Inputs

2.1 Primitives invoked

- **P08 (Substrate scale ℓ_{ED})** — supplies the substrate-level length scale.
- **P10 (Rule-type primitive)** — supplies V1 kernel rule-type.
- **P11 (Commitment-irreversibility)** — supplies V1 retardation.

2.2 Upstream Dependencies

- **I-027:** Newton’s law $G = c^3 \ell_{\text{ED}}^2 / \hbar$ at substrate level (Paper_027).
- **I-029:** Transition acceleration $a_0 = cH_0 / (2\pi)$ (Paper_029).
- **I-089:** V1 finite-width retarded kernel — long-range behavior at scales $r \gg \ell_{\text{ED}}$.
- **I-RQM-hbar:** $\hbar$ as participation quantum.
- **I-Newton:** Standard Newton’s law machinery.

3 Derivation

3.1 Newton’s law at substrate level

Paper_027 derives Newton’s law $a = GM/r^2$ at substrate level with the gravitational constant given by

$$G = \frac{c^3 \ell_{\text{ED}}^2}{\hbar},$$

where ℓ_{ED} is the P08 substrate scale.

The substrate-level mechanism: V1 long-range kernel content produces a $1/r^2$ acceleration at scales $r \gg \ell_{\text{ED}}$ (Paper_089); the proportionality constant involves ℓ_{ED}^2 through the substrate participation-budget scaling.

3.2 Empirical matching to Newton’s law

Empirically, $G \approx 6.674 \times 10^{-11} \text{ m}^3/(\text{kg s}^2)$.

The Planck length is defined as

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35} \text{ m}.$$

Substituting $G = c^3 \ell_{\text{ED}}^2 / \hbar$ from Paper_027:

$$\ell_P^2 = \frac{\hbar}{c^3} \cdot \frac{c^3 \ell_{\text{ED}}^2}{\hbar} = \ell_{\text{ED}}^2.$$

Therefore $\ell_{\text{ED}} = \ell_P$. The substrate scale equals the Planck length.

3.3 Why this identification is forced

Three independent constraints jointly force $\ell_{\text{ED}} = \ell_P$:

Constraint 1: Newton-recovery. The substrate-gravity arc must reproduce Newton’s law at appropriate scales. Paper_027’s derivation gives G in terms of ℓ_{ED} .

Constraint 2: V1 long-range behavior. V1 kernel (Paper_089) has finite second moment $\ell_{V1}^2 \sim \ell_{ED}^2$; long-range $1/r^2$ behavior emerges at scales $r \gg \ell_{ED}$.

Constraint 3: Empirical G value. G has been measured to high precision; combined with Constraint 1, this fixes $\ell_{ED}^2 = \hbar G / c^3 = \ell_P^2$.

The combination is forced: any deviation $\ell_{ED} \neq \ell_P$ would produce a different G , contradicting empirical matching. The Planck identification is therefore the **unique** substrate-scale consistent with substrate gravity + Newton's law.

3.4 Why ℓ_P , not some other Planck-class scale

Multiple Planck-class scales exist in standard physics:

- $\ell_P = \sqrt{\hbar G / c^3} \approx 1.6 \times 10^{-35}$ m.
- Reduced Planck length $\ell_P / \sqrt{8\pi}$.
- Planck mass-corresponding lengths (electroweak scale, GUT scale, etc.).

The substrate framework picks out **specifically** $\ell_P = \sqrt{\hbar G / c^3}$ because this is the scale that appears directly in $G = c^3 \ell_{ED}^2 / \hbar$ from Paper_027. Other conventions (reduced Planck, etc.) differ by $O(1)$ factors absorbed into the substrate normalization but do not change the identification's structural content.

3.5 Consequences of the identification

With $\ell_{ED} = \ell_P$:

- **V1 width:** $\ell_{V1} \sim \ell_P$ (Paper_089 numerical).
- **V5 cutoff:** $\omega_c = c / \ell_P$ (Paper_040).
- **Substrate UV cutoff:** ℓ_P (Paper_RQM_T9 primitive-level UV finiteness).
- **BH-horizon scale:** $r_H = 2GM / c^2$ in Planck units.
- **Cosmological scale:** $R_H = c / H_0$, related to substrate via $\ell_P \cdot H_0 \cdot t_{\text{universe-age}}$ structural factors.

All these scales share the substrate origin at ℓ_P . The Planck identification is the substrate framework's anchor for the fundamental-constant system.

3.6 What is INHERITED vs FORM-FORCED

FORM-FORCED:

- Identification $\ell_{ED} = \ell_P$ structurally.
- Substrate-scale is the **same** as the Planck-length scale (not coincidentally; structurally).
- Universality of ℓ_P across all gravity-arc substrate quantities.

INHERITED:

- Numerical value of ℓ_P (via $\hbar$, G , c empirical values).
- Numerical value of G itself.
- Numerical value of $\hbar$.

In the H1-dominant regime (Paper_RQM_ArcM_H1 + Paper_RQM_hbar), these fundamental constants are structurally identified but numerically inherited.

3.7 Future derivation of ℓ_P

A substrate-level first-principles derivation of ℓ_P value would require:

- Substrate derivation of $\mathcal{B}_{P04}$ bandwidth-density (Paper_RQM_hbar Route).

- Substrate derivation of substrate-cell volume (currently inherited via ℓ_P).
- Composition to give absolute ℓ_P value.

This is OPEN — analogous to Paper_ED_SC_4.2 Route A.

4 Audit Table

#	Step	Label	Notes
1	P08 substrate scale ℓ_{ED}	P	Primitive.
2	P10 rule-type	P	Primitive.
3	Newton's law derived $G = c^3 \ell_{ED}^2 / \hbar$	I	Paper_027.
4	V1 long-range kernel structure	I	Paper_089.
5	Empirical G value	I	Empirical.
6	Planck length defined $\ell_P = \sqrt{\hbar G / c^3}$	I	Standard physics.
7	Substitution: $\ell_P^2 = \hbar(c^3 \ell_{ED}^2 / \hbar) / c^3 = \ell_{ED}^2$	D	Direct algebra.
8	$\ell_{ED} = \ell_P$ identification	I	Dimensional rearrangement of $G, \hbar, c$; not independent substrate derivation.
9	Three-constraint forcing argument	D	Composition of Newton-recovery + V1 + empirical G .
10	ℓ_P as universal substrate scale	D	From step 8.
11	Substrate UV cutoff at ℓ_P	I	Paper_RQM_T9, Paper_040.
12	ℓ_P numerical value	I	Empirical.
13	Verdict M3	A→position	Per Paper_095. Form-FORCED Newton recovery + value-INHERITED Planck identification.

5 Falsification Criteria

- **F1:** Substrate-level derivation of $\ell_{ED} \neq \ell_P$ — refutes the Newton-recovery identification.
- **F2:** Empirical refutation of Newton's law $a = GM/r^2$ in the appropriate substrate-gravity regime — refutes Paper_027 (and propagates here).
- **F3:** V1 long-range behavior shown inconsistent with $1/r^2$ — refutes Paper_089 (propagates).
- **F4:** Substrate derivation of ℓ_P from first principles (upgrade, not refutation).

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton interpretation of QM; Wolfram Ruliad / hypergraph-rewriting; causal-set program, Sorkin et al.), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

This paper supplies the substrate-level identification $\ell_{ED} = \ell_P$ as a forced consequence of substrate-gravity Newton-recovery.

Relationship to other papers:

- **Paper_027:** supplies the substrate-derived $G = c^3 \ell_{ED}^2 / \hbar$.
- **Paper_089:** V1 long-range structure.
- **Paper_RQM_T9:** UV cutoff at ℓ_P .
- **Paper_RQM_hbar:** companion identification for $\hbar$.

Numerical content INHERITED. ℓ_P , G , $\hbar$ values. **Form FORCED.** $\ell_{ED} = \ell_P$ structural identification.

Verdict: **M3.**